5.1 Approximating and Computing Area

Summation Notation

Summation notation is a standard notation for writing sums in compact form. The sum of numbers $a_m, \ldots, a_n$ $(m \leq n)$ is denoted

$$\sum_{j=m}^{n} a_j = a_m + a_{m+1} + \cdots + a_n$$

The Greek letter $\sum$ (capital sigma) stands for "sum," and the notation $\sum_{j=m}^{n}$ tells us to start the summation at $j = m$ and end it at $j = n$. For example,

$$\sum_{j=1}^{5} j^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

Ex: $\sum_{k=4}^{6} (k^2 - 2k) = ?$

$$= (\cancel{4^2}^{16} - \cancel{2(4)}^{8}) + (\cancel{5^2}^{25} - \cancel{2(5)}^{10}) + (\cancel{6^2}^{36} - \cancel{2(6)}^{12})$$

$$= 8 + 15 + 24$$

$$= \boxed{47}$$ ✓

$$\sum_{j=1}^{5} j^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

In this summation, the jth term is $a_j = j^2$. We refer to j^2 as the general term. The letter j is called the summation index. It is also referred to as a **dummy variable** because any other letter can be used instead. For example,

$$\sum_{k=4}^{6} (k^2 - 2k) = \overbrace{(4^2 - 2(4))}^{k=4} + \overbrace{(5^2 - 2(5))}^{k=5} + \overbrace{(6^2 - 2(6))}^{k=6} = 47$$

$$\sum_{m=7}^{9} 1 = 1 + 1 + 1 = 3 \qquad \text{(because } a_7 = a_8 = a_9 = 1\text{)}$$

Linearity of Summations

- $\displaystyle\sum_{j=m}^{n}(a_j + b_j) = \sum_{j=m}^{n} a_j + \sum_{j=m}^{n} b_j$
- $\displaystyle\sum_{j=m}^{n} C a_j = C \sum_{j=m}^{n} a_j \qquad (C \text{ any constant})$
- $\displaystyle\sum_{j=1}^{n} C = nC \qquad (C \text{ any constant and } n \geq 1)$

For example,

$$\sum_{j=3}^{5}(j^2+j)=(3^2+3)+(4^2+4)+(5^2+5)=62$$

can also be expressed as

$$\sum_{j=3}^{5}j^2+\sum_{j=3}^{5}j=(3^2+4^2+5^2)+(3+4+5)=50+12=62$$

The linearity properties can be used to write a single summation as a linear combination of several summations. For example,

$$\begin{aligned}\sum_{k=0}^{100}(7k^2-4k+9) &= \sum_{k=0}^{100}7k^2+\sum_{k=0}^{100}(-4k)+\sum_{k=0}^{100}9 \\ &= 7\sum_{k=0}^{100}k^2-4\sum_{k=0}^{100}k+9\sum_{k=0}^{100}1\end{aligned}$$

Power Sums

$$\sum_{j=1}^{N} j = 1 + 2 + \cdots + N = \frac{N(N+1)}{2} = \frac{N^2}{2} + \frac{N}{2} \tag{3}$$

$$\sum_{j=1}^{N} j^2 = 1^2 + 2^2 + \cdots + N^2 = \frac{N(N+1)(2N+1)}{6} = \frac{N^3}{3} + \frac{N^2}{2} + \tag{4}$$

$$\sum_{j=1}^{N} j^3 = 1^3 + 2^3 + \cdots + N^3 = \frac{N^2(N+1)^2}{4} = \frac{N^4}{4} + \frac{N^3}{2} + \frac{N^2}{4} \tag{5}$$

5.1 Approximating and Computing Area

Why might we be interested in the area under a graph? Consider an object moving in a straight line with *constant positive velocity* v. The distance traveled over a time interval $[t_1, t_2]$ is equal to $v\Delta t$, where $\Delta t = (t_2 - t_1)$ is the time elapsed. This is the well-known formula

$$\text{distance traveled} = \underbrace{\text{velocity} \times \text{time elapsed}}_{v\Delta t} \qquad \textbf{1}$$

Because v is constant, the graph of velocity is a horizontal line (Figure 1) and $v\Delta t$ is equal to the area of the rectangular region under the graph of velocity over $[t_1, t_2]$. So we can write Eq. (1) as

$$\text{distance traveled} = \text{area under the graph of velocity over } [t_1, t_2] \qquad \textbf{2}$$

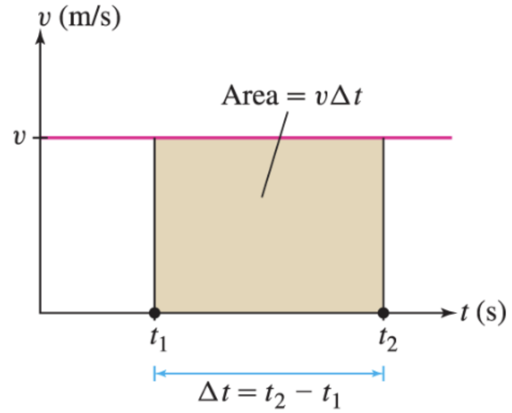


FIGURE 1 The rectangle has area $v\Delta t$, which is equal to the distance traveled.

Our strategy when velocity changes continuously (Figure 3) is to *approximate* the area under the graph by a sum of areas of rectangles. We can continually improve the approximation by using thinner and thinner rectangles, and then take a limit to obtain an exact value of distance traveled. This idea leads to the concept of an integral.

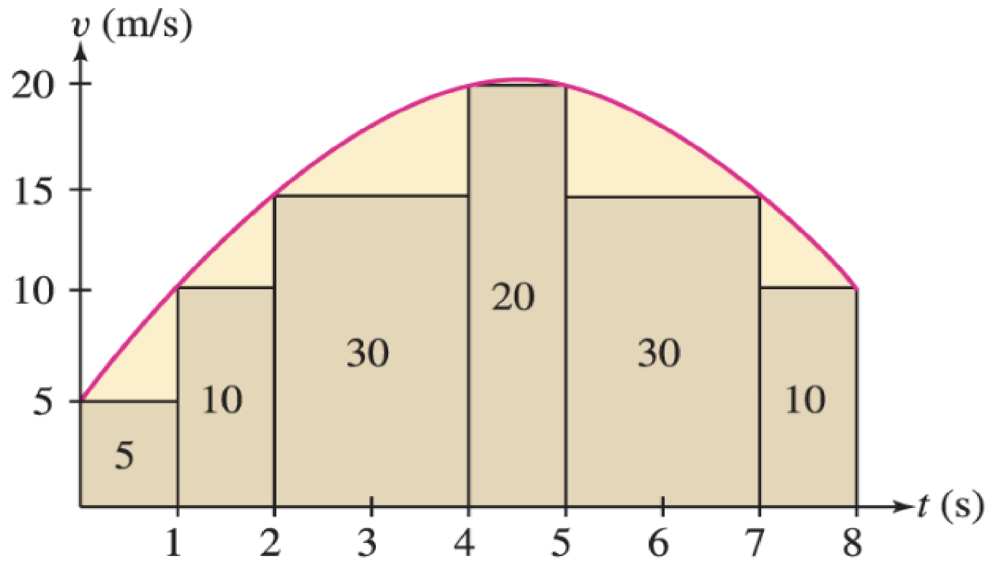


FIGURE 3 Distance traveled is equal to the area under the graph. It is *approximated* by the sum of the areas of the rectangles.

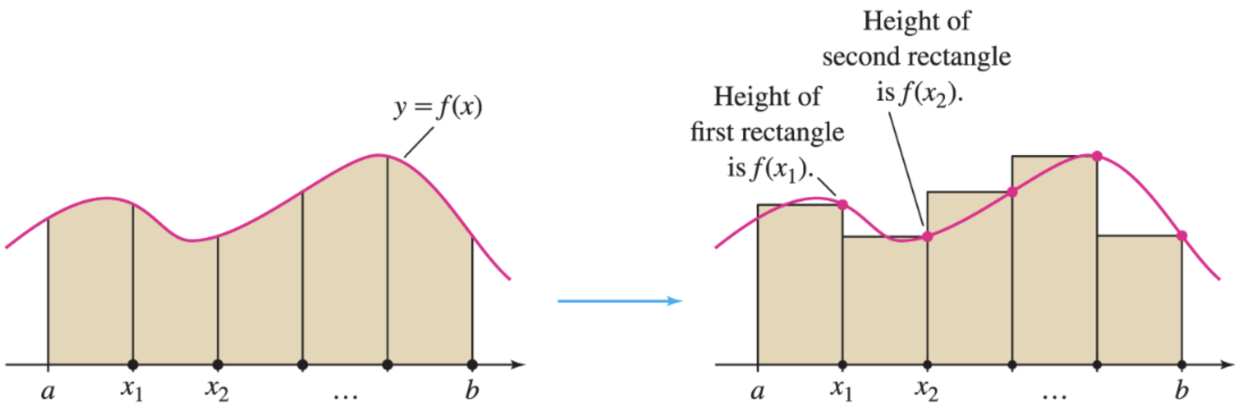
Approximating Area by Rectangles

Our goal is to compute the area under the graph of a function f. In this section, we assume that f is continuous and *positive*, so that the graph of f lies above the x-axis (Figure 4). The first step is to approximate the area using rectangles.

To begin, choose a Approximating Area by Rectangles $[a, b]$ into N subintervals of equal width, as in Figure 4(A). The full interval $[a, b]$ has width $b - a$, so each subinterval has width $\Delta x = (b - a)/N$. The right endpoints of the subintervals are

$$x_1 = a + \Delta x,\ x_2 = a + 2\Delta x, \ldots,\ x_{N-1} = a + (N-1)\Delta x,\ x_N = a +$$

Note that the last right endpoint is $x_N = b$ because $a + N\Delta x = a + N((b - a)/N) = b$. The general term x_j can be expressed as $x_j = a + j\Delta x$, indicating that it is obtained by adding j intervals of width Δx to a. Next, as in Figure 4(B), above each subinterval construct a rectangle whose height is the value of $f(x)$ at the *right endpoint* of the subinterval.



(A) Divide $[a, b]$ into N subintervals, each of width Δx.

(B) Construct right-endpoint rectangles.

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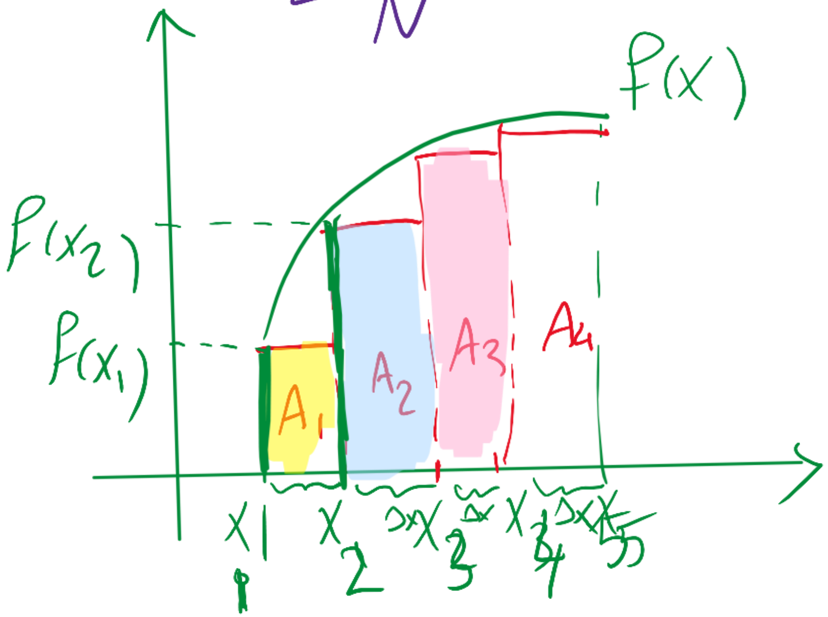
FIGURE 4

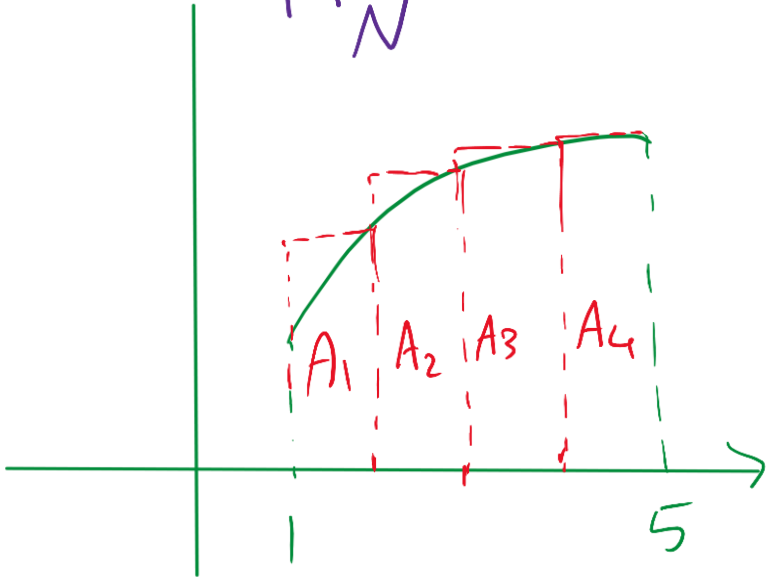
Recall the two-step procedure for finding the slope of the tangent line (the derivative): First approximate the slope using secant lines and then compute the limit of these approximations. In integral calculus, there are also two steps:

- *First, approximate the area under the graph using rectangles.*
- *Then compute the exact area (the integral) as the limit of these approximations.*

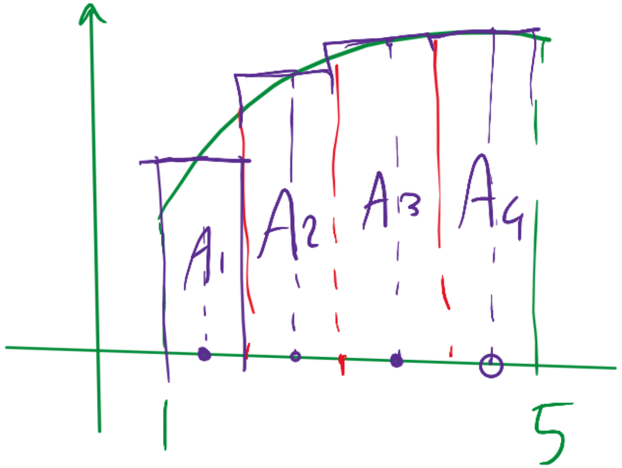
L_N



R_N



M_N



$A \approx A_1 + A_2 + A_3 + A_4$

$A \approx A_1 + A_2 + A_3 + A_4$

$M = A_1 + A_2 + A_3 + A_4$

R

$$\Delta x \cdot f(x_1) + \Delta x \cdot f(x_2) + \Delta x \cdot f(x_4) + \cdots \Delta x \cdot f(x_n)$$

$$\Delta x \left(f(x_1) + f(x_2) + f(x_3) + \cdots f(x_n) \right)$$

$$\boxed{R_N = \Delta x \sum_{i=1}^{n} f(x_i)}$$

The sum of the areas of these rectangles provides an *approximation* to the area under the graph. The first rectangle has base Δx and height $f(x_1)$, so its area is $f(x_1)\Delta x$. Similarly, the second rectangle has height $f(x_2)$ and area $f(x_2)\Delta x$, and so on. The sum of the areas of the rectangles is denoted R_N and is called the **Nth right-endpoint approximation**:

$$R_N = f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_N)\Delta x$$

Factoring out Δx, we obtain the formula

$$R_N = \Delta x\left(f(x_1) + f(x_2) + \cdots + f(x_N)\right)$$

To summarize:

$$\begin{aligned} a &= left\ endpoint\ of\ interval\ [a, b] \\ b &= right\ endpoint\ of\ interval\ [a, b] \\ N &= number\ of\ subintervals\ in\ [a, b] \\ \Delta x &= \boxed{\frac{b-a}{N}} \end{aligned}$$

100% in Exam

EXAMPLE 1

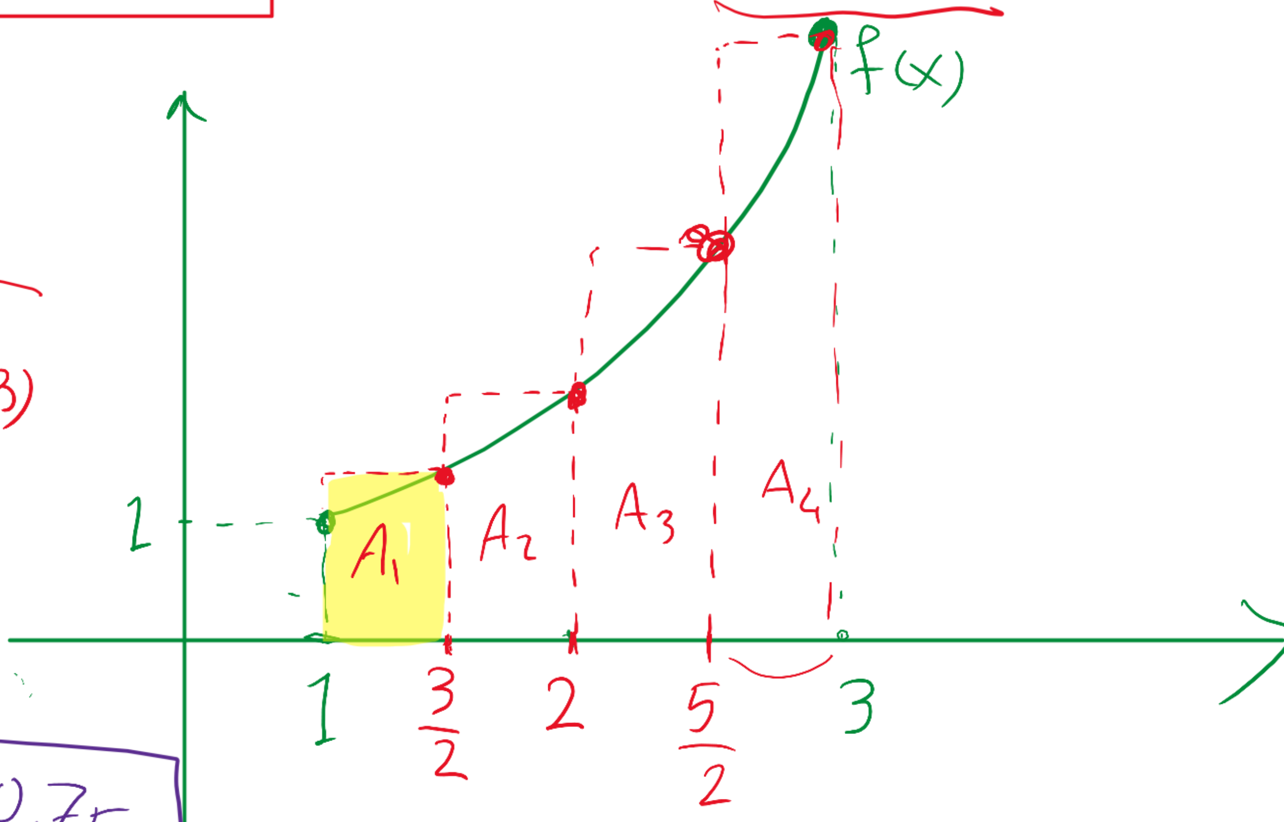
Calculate R_4 and R_6 for $\boxed{f(x) = x^2}$ on the interval $[1, 3]$.

$$\Delta x = \frac{b-a}{N} = \frac{3-1}{4} = \frac{2}{4} = \boxed{\frac{1}{2}}$$

$$\overbrace{\frac{1}{2} \cdot f\left(\frac{3}{2}\right)}^{A_1} + \overbrace{\frac{1}{2} \cdot f(2)}^{A_2} + \overbrace{\frac{1}{2} \cdot f\left(\frac{5}{2}\right)}^{A_3} + \overbrace{\frac{1}{2} \cdot f(3)}^{A_4}$$

$$\frac{1}{2}\left(f\left(\frac{3}{2}\right) + f(2) + f\left(\frac{5}{2}\right) + f(3)\right)$$

$$\frac{1}{2}\left(\underbrace{\cancel{\left(\frac{3}{2}\right)^2}}_{\frac{9}{4}} + \underbrace{\cancel{(2)^2}}_{4} + \underbrace{\left(\frac{5}{2}\right)^2}_{\frac{25}{4}} + \underbrace{\cancel{3^2}}_{9}\right) = \frac{43}{4} \approx \boxed{10.75}$$



Solution

***Step 1*. Determine Δx and the right endpoints.**

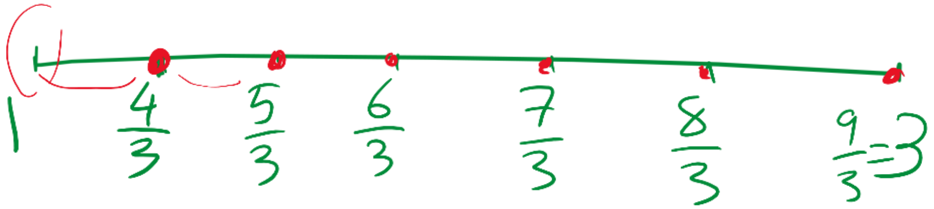
To calculate R_4, divide $[1, 3]$ into four subintervals of width $\Delta x = \frac{3-1}{4} = \frac{1}{2}$. The right endpoints are the numbers $x_j = a + j\Delta x = 1 + j\left(\frac{1}{2}\right)$ for $j = 1, 2, 3, 4$. They are spaced at intervals of $\frac{1}{2}$ beginning at $\frac{3}{2}$, so, as we see in Figure 5(A), the right endpoints are $\frac{3}{2}, \frac{4}{2}, \frac{5}{2}, \frac{6}{2}$.

***Step 2*. Calculate Δx times the sum of function values.**

R_4 is Δx times the sum of the function values at the right endpoints:

$$\begin{aligned} R_4 &= \frac{1}{2}\left(f\left(\frac{3}{2}\right)+f\left(\frac{4}{2}\right)+f\left(\frac{5}{2}\right)+f\left(\frac{6}{2}\right)\right) \\ &= \frac{1}{2}\left(\left(\frac{3}{2}\right)^2+\left(\frac{4}{2}\right)^2+\left(\frac{5}{2}\right)^2+\left(\frac{6}{2}\right)^2\right)= \frac{43}{4} = 10.75 \end{aligned}$$

$1+\frac{1}{3}=\frac{3}{3}+\frac{1}{3}=\frac{4}{3}$



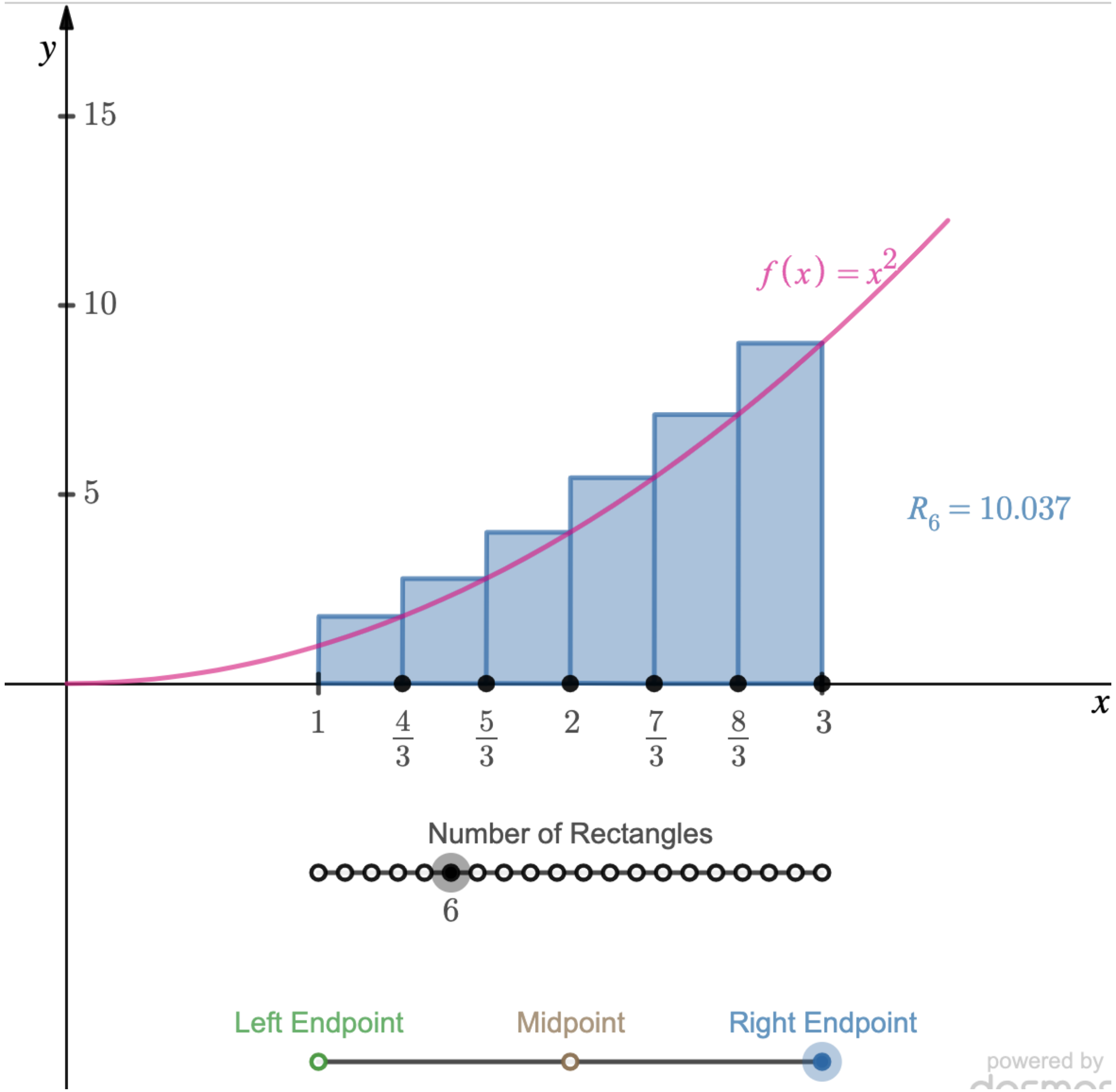
R_6 is similar: $\Delta x = \frac{3-1}{6} = \frac{1}{3}$, and the right endpoints are spaced at intervals of $\frac{1}{3}$ beginning at $\frac{4}{3}$ and ending at 3, as in Figure 5(B). Thus,

$$\begin{aligned}
R_6 &= \frac{1}{3}\left(f\left(\frac{4}{3}\right)+f\left(\frac{5}{3}\right)+f\left(\frac{6}{3}\right)+f\left(\frac{7}{3}\right)+f\left(\frac{8}{3}\right)+f\left(\frac{9}{3}\right)\right) \\
&= \frac{1}{3}\left(\frac{16}{9}+\frac{25}{9}+\frac{36}{9}+\frac{49}{9}+\frac{64}{9}+\frac{81}{9}\right) = \frac{271}{27} \approx \boxed{10.037}
\end{aligned}$$

$\approx$

$R_4 \approx 10.75$

$R_6 = 10.037$



y
15
10
5
x
1
4/3
5/3
2
7/3
8/3
3
f(x) = x^2
R_6 = 10.037
Number of Rectangles
6
Left Endpoint
Midpoint
Right Endpoint
powered by

The R_4 and R_6 values in the previous example are clearly overestimates of the area under $y = x^2$ between 1 and 3, although the approximation improves going from R_4 to R_6. Later in the section, we show how to obtain a general expression for R_N. With that, we can then take the limit as $N \to \infty$ to obtain an exact value of $8\,\frac{2}{3}$ as the area (see Exercise 55).

It is convenient to use summation notation when working with area approximations. For example, R_N is a sum with general term $f(x_j)$:

$$R_N = \Delta x \left[f(x_1) + f(x_2) + \cdots + f(x_N) \right]$$

The summation extends from $j = 1$ to $j = N$, so we can write R_N concisely as

R_4

R_6

$N \to \infty$

$$\boxed{R_N = \Delta x \sum_{j=1}^{N} f(x_j)}$$

We shall make use of two other rectangular approximations to area: the left-endpoint and the midpoint approximations. Divide $[a, b]$ into N subintervals as before. In the **left-endpoint approximation** L_N, the heights of the rectangles are the values of $f(x)$ at the left endpoints [Figure 6(A)]. These left endpoints are

$$x_0 = a, \quad x_1 = a + \Delta x, \quad x_2 = a + 2\Delta x, \quad \ldots, \quad x_{N-1} = a + (N-1)\Delta x$$

and the sum of the areas of the left-endpoint rectangles is

$$L_N = \Delta x(f(x_0) + f(x_1) + f(x_2) + \cdots + f(x_N - 1))$$

REMINDER

$$\Delta x = \frac{b-a}{N}$$

Note that both R_N and L_N have general term $f(x_j)$, but the sum for L_N runs from $j = 0$ to $j = N - 1$ rather than from $j = 1$ to $j = N$:

$$\boxed{L_N = \Delta x \sum_{j=0}^{N-1} f(x_j)}$$

In the **midpoint approximation** M_N, the heights of the rectangles are the values of $f(x)$ at the midpoints of the subintervals rather than at the endpoints. As we see in Figure 6(B), the midpoints are

$$\frac{x_0 + x_1}{2}, \quad \frac{x_1 + x_2}{2}, \quad \dots, \quad \frac{x_{N-1} + x_N}{2}$$

The sum of the areas of the midpoint rectangles is

$$M_N = \Delta x \left(f\left(\frac{x_0 + x_1}{2} \right) + f\left(\frac{x_1 + x_2}{2} \right) + \cdots + f\left(\frac{x_{N-1} + x_N}{2} \right) \right)$$

In summation notation,

$$M_N = \Delta x \sum_{j=0}^{N-1} f\left(\frac{x_j + x_{j+1}}{2} \right)$$

EXAMPLE 2

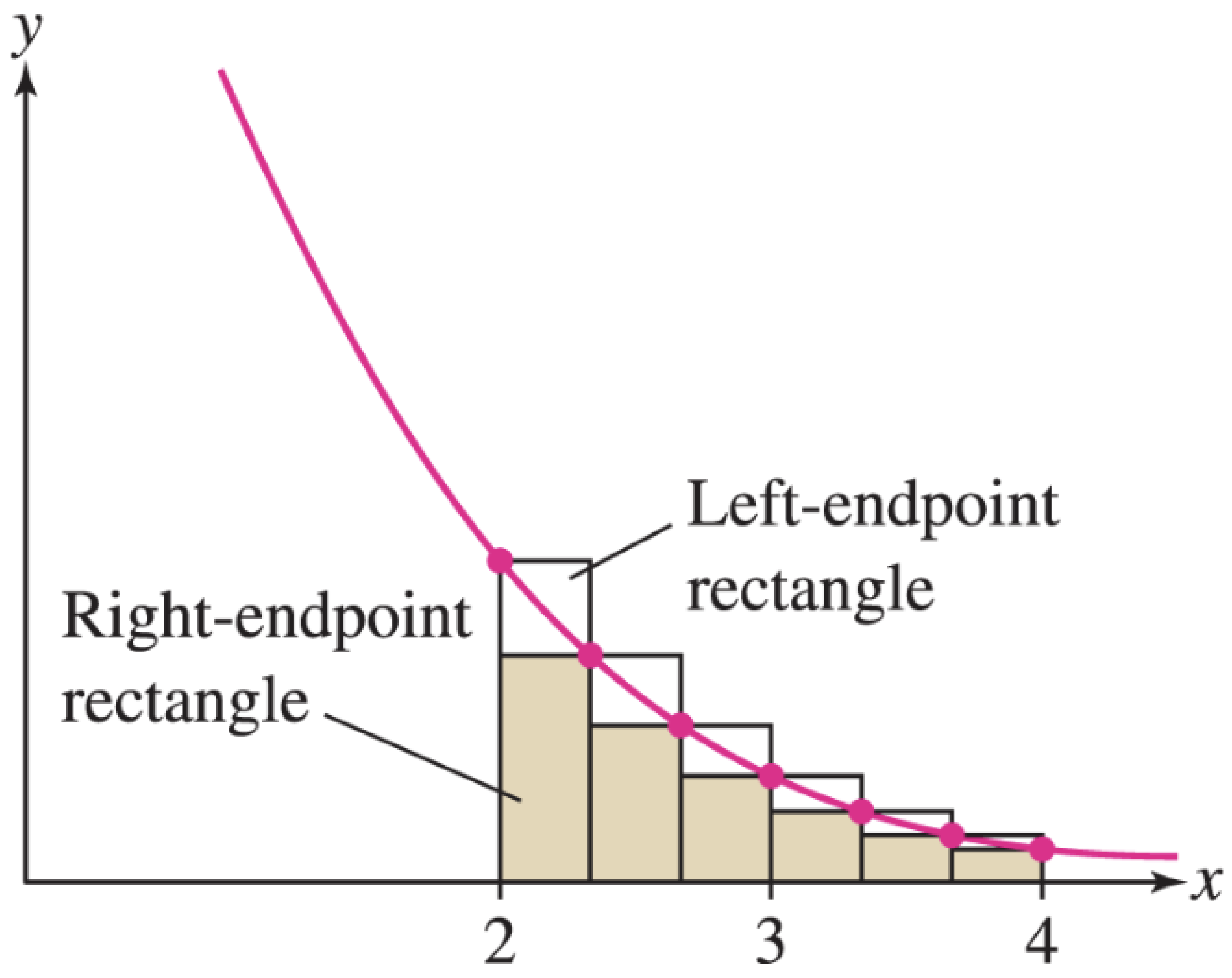
Calculate R_6, L_6, and M_6 for $f(x) = x^{-1}$ on $[2, 4]$.

Solution

In this case, $\Delta x = (b-a)/N = (4-2)/6 = \frac{1}{3}$. For the six intervals, the right endpoints are $\frac{7}{3}, \frac{8}{3}, 3, \frac{10}{3}, \frac{11}{3}$, and 4, and the left endpoints are $2, \frac{7}{3}, \frac{8}{3}, 3, \frac{10}{3}$ and $\frac{11}{3}$.

Therefore (Figure 7),

$$\begin{aligned} R_6 &= \frac{1}{3}\left(f\left(\frac{7}{3}\right)+f\left(\frac{8}{3}\right)+f(3)+f\left(\frac{10}{3}\right)+f\left(\frac{11}{3}\right)+f(4)\right) \\ &= \frac{1}{3}\left(\frac{3}{7}+\frac{3}{8}+\frac{1}{3}+\frac{3}{10}+\frac{3}{11}+\frac{1}{4}\right) \approx 0.653 \\ L_6 &= \frac{1}{3}\left(f(2)+f\left(\frac{7}{3}\right)+f\left(\frac{8}{3}\right)+f(3)+f\left(\frac{10}{3}\right)+f\left(\frac{11}{3}\right)\right) \\ &= \frac{1}{3}\left(\frac{1}{2}+\frac{3}{7}+\frac{3}{8}+\frac{1}{3}+\frac{3}{10}+\frac{3}{11}\right) \approx 0.737 \end{aligned}$$

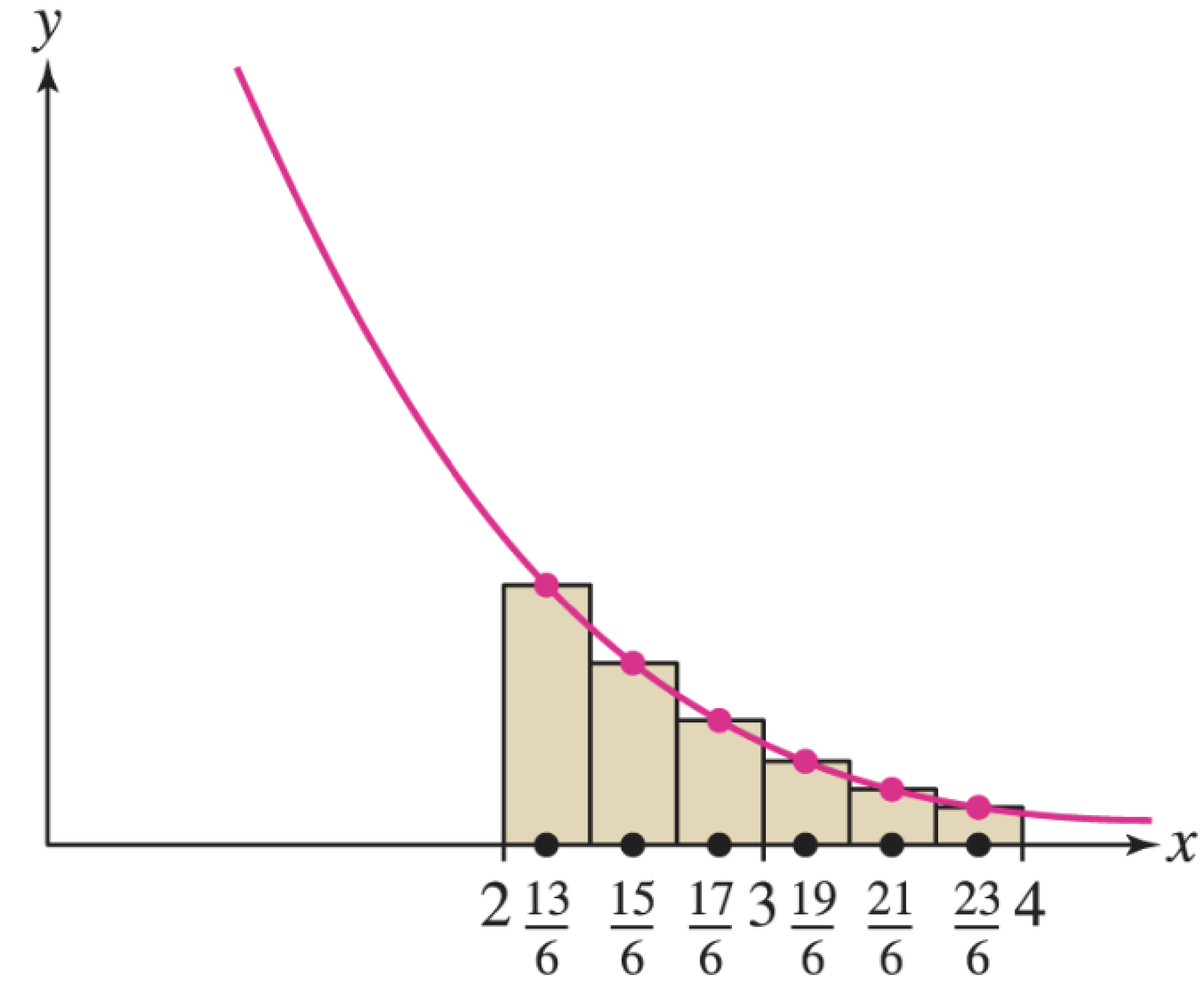

y
Left-endpoint
rectangle
Right-endpoint
rectangle
2
3
4
x

The general term in M_6 is

$$f\left(\frac{x_j + x_{j+1}}{2}\right)$$

In this case, the midpoints are $\frac{13}{6}$, $\frac{15}{6}$, $\frac{17}{6}$, $\frac{19}{6}$, $\frac{21}{6}$ and $\frac{23}{6}$. Summing from $j = 0$ to 5, we obtain (Figure 8)

$$\begin{aligned} M_6 &= \frac{1}{3}\left(f\left(\frac{13}{6}\right)+f\left(\frac{15}{6}\right)+f\left(\frac{17}{6}\right)+f\left(\frac{19}{6}\right)+f\left(\frac{21}{6}\right)+f\left(\frac{23}{6}\right)\right) \\ &= \frac{1}{3}\left(\frac{6}{13}+\frac{6}{15}+\frac{6}{17}+\frac{6}{19}+\frac{6}{21}+\frac{6}{23}\right) \approx 0.692 \end{aligned}$$


y
x
2
13/6
15/6
17/6
3
19/6
21/6
23/6
4

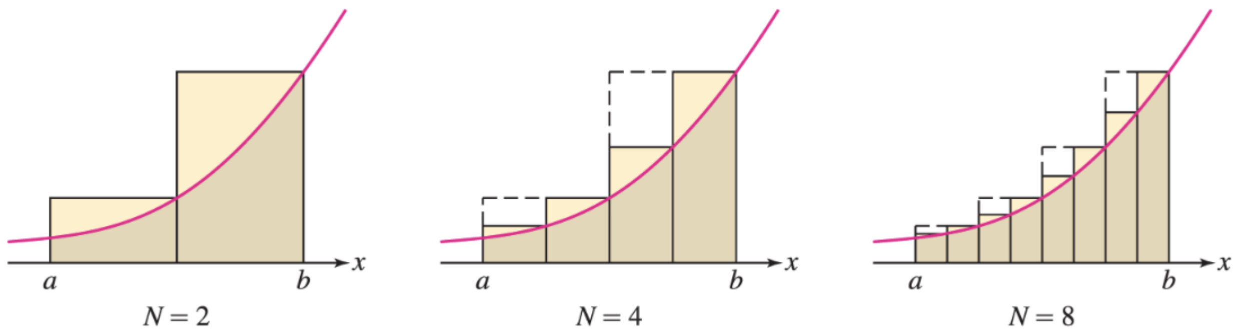


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FIGURE 10 The error decreases as we use more rectangles.

THEOREM 1

If f is continuous on $[a, b]$, then the endpoint and midpoint approximations approach one and the same limit as $N \to \infty$. In other words, there is a value L such that

$$\lim_{N\to\infty} R_N = \lim_{N\to\infty} L_N = \lim_{N\to\infty} M_N = L$$

If $f(x) \geq 0$ on $[a, b]$, we define the area under the graph over $[a, b]$ to be L.

EXAMPLE 3

Find the area A under the graph of $f(x) = x$ over $[0, 4]$ in three ways:

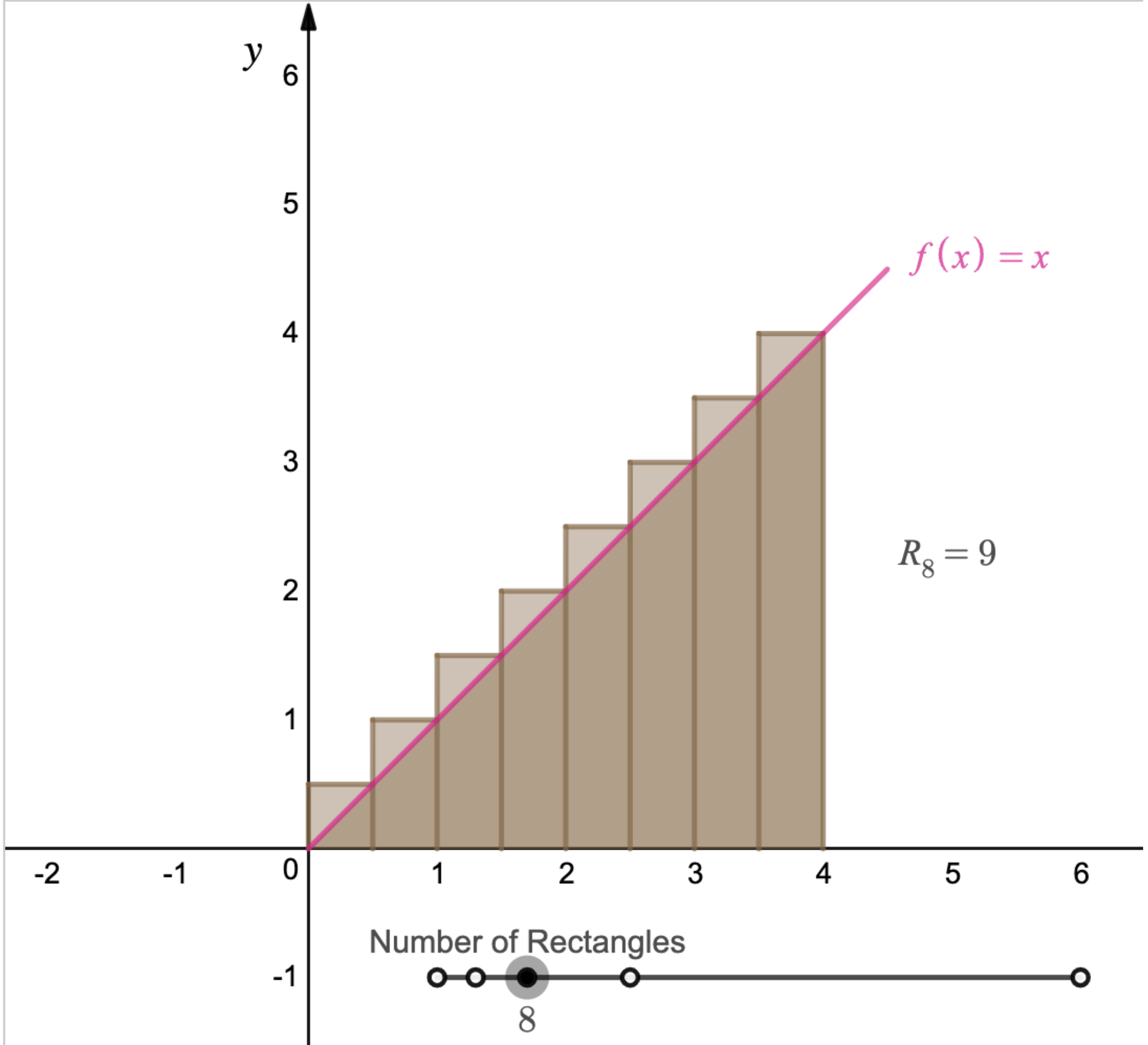
a. Using geometry

b. $\lim\limits_{N \to \infty} R_N$

c. $\lim\limits_{N \to \infty} L_N$

Solution

The region under the graph is a right triangle with base $b = 4$ and height $h = 4$ (Figure 11).



a. By geometry, $A = \frac{1}{2}bh = \left(\frac{1}{2}\right)(4)(4) = 8$.

b. We compute this area again as a limit. Since $\Delta x = (b - a)/N = 4/N$ and $f(x) = x$,

$$\begin{aligned} f(x_j) &= f(a + j\Delta x) = f\left(0 + j\left(\frac{4}{N}\right)\right) = \frac{4j}{N} \\ R_N &= \Delta x \sum_{j=1}^{N} f(x_j) = \frac{4}{N} \sum_{j=1}^{N} \frac{4j}{N} = \frac{16}{N^2} \sum_{j=1}^{N} j \end{aligned}$$

In the last equality, we factored out $4/N$ from the sum. This is valid because $4/N$ is a constant that does not depend on j. Now use formula (3):

$$R_N = \frac{16}{N^2} \sum_{j=1}^{N} j = \frac{16}{N^2} \underbrace{\left(\frac{N(N+1)}{2}\right)}_{\text{Formula for power sum}} = \frac{8}{N^2}\left(N^2 + N\right) = 8 + \frac{8}{N}$$

The second term $8/N$ tends to zero as N approaches ∞, so

$$A = \lim_{N\to\infty} R_N = \lim_{N\to\infty} \left(8 + \frac{8}{N}\right) = 8$$

REMINDER

$$R_N = \Delta x \sum_{j=1}^{N} f(x_j)$$

$$L_N = \Delta x \sum_{j=0}^{N-1} f(x_j)$$

$$\Delta x = \frac{b-a}{N}$$

$$x_j = a + j\Delta x$$

c. The left-endpoint approximation is similar, but the sum begins at $j = 0$ and ends at $j = N - 1$:

$$L_N = \frac{16}{N^2} \sum_{j=0}^{N-1} j = \frac{16}{N^2} \sum_{j=1}^{N-1} j = \frac{16}{N^2} \left(\frac{(N-1)\,N}{2} \right) = 8 - \frac{8}{N}$$

Note in the second step that we replaced the sum beginning at $j = 0$ with a sum beginning at $j = 1$. This is valid because the term for $j = 0$ is zero and may be dropped. Again, we find that $A = \lim_{N \to \infty} L_N = \lim_{N \to \infty} (8 - 8/N) = 8$.

5.1 SUMMARY

- Approximations to the area under the graph of f over the interval $[a, b]$ $\left(\Delta x = \dfrac{b-a}{N}, x_j = a + j\Delta x\right)$:

$$\begin{aligned}
R_N &= \Delta x \sum_{j=1}^{N} f(x_j) = \Delta x(f(x_1) + f(x_2) + \cdots + f(x_N)) \\
L_N &= \Delta x \sum_{j=0}^{N-1} f(x_j) = \Delta x(f(x_0) + f(x_1) + \cdots + f(x_{N-1})) \\
M_N &= \Delta x \sum_{j=0}^{N-1} f\left(\frac{x_j + x_{j+1}}{2}\right) \\
&= \Delta x\left(f\left(\frac{x_0 + x_1}{2}\right) + \cdots + f\left(\frac{x_{N-1} + x_N}{2}\right)\right)
\end{aligned}$$

Power Sums

$$\sum_{j=1}^{N} j = \frac{N(N+1)}{2} = \frac{N^2}{2} + \frac{N}{2}$$

$$\sum_{j=1}^{N} j^2 = \frac{N(N+1)(2N+1)}{6} = \frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6}$$

$$\sum_{j=1}^{N} j^3 = \frac{N^2(N+1)^2}{4} = \frac{N^4}{4} + \frac{N^3}{2} + \frac{N^2}{4}$$